YZV475E · Data Visualization
2025 – 2026 Spring

# Term Project Proposal

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## 1. Group Information

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| **Group Name** | DataVIZZ |
| **Member 1** | Nurettin Macit 150220329 macit22@itu.edu.tr |
| **Member 2** | Mehmet Arda Öncel 150230312 oncel23@itu.edu.tr |
| **Member 3** | Yusuf Oğuz 150220322 oguz22@itu.edu.tr |

## 2. Dataset

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| **Dataset Name** | NBA Players & Teams Statistics (1947–2024) |
| **Source / URL** | https://www.kaggle.com/datasets/sumitrodatta/nba-aba-baa-stats |
| **License** | Public Domain |
| **Size** | 33,278 rows × 32 columns<br>11 MB |

### Dataset Description

**This dataset is compiled from Basketball Reference and covers the NBA, ABA, and BAA leagues from the 1947 to 2024 seasons. It consists of 22 CSV files containing per-game player statistics, advanced metrics, draft history, All-Star selections, and team summaries. Data is structured annually at both the player and team level, making it highly suitable for time series, distribution, and comparative visualizations.**

## 3. Motivation & Research Questions

**The NBA is one of the most data-rich sports leagues in the world, having undergone fundamental tactical and physical transformations since 1947. We chose this dataset because it offers the opportunity to visually explore nearly 80 years of historical data and uncover meaningful patterns in how the game has evolved.**

**Research Questions:**
**1. How has the three-point revolution transformed NBA gameplay over the decades?**

**2. Is there a meaningful relationship between players' physical attributes and their performance statistics?**

**3. Which countries and U.S. states produce the most NBA players?**

**4. How do the statistical profiles of different positions differ?**

**5. How does player performance develop throughout a career, and at what age do players typically reach their peak?**

## 4. Planned Visualizations

Your dashboard must include at least 3 of the following visual types. Check all that apply and describe your plan for each selected type.

| ✓ | Visual Type | Brief Description of Your Plan |
|---|---|---|
| + | Time Series Plot | Annual trend of 3-point attempt averages, scoring averages, and pace from 1947 to 2024, visualizing how the style of play has shifted over time. |
| + | Geospatial Figure | World map showing the number of NBA players and average performance by birth country; U.S. state-level choropleth map showing player distribution by state. |
| ☐ | Graph Visualization | |
| + | Distribution Plot | Comparative histograms and box plots of player age, points per game and height across different eras and positions. |
| ☐ | Hierarchical Data | |
| + | Scatter Plot | Relationship between points per game and Win Shares, color-coded by position and era to reveal performance patterns. |
| + | Heatmap | Position × statistics correlation matrix showing how key metrics vary across playing positions. |

## 5. Tools & Technologies

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| **Programming Language(s)** | Python |
| **Visualization Libraries** | Plotly, Seaborn, Matplotlib |
| **Dashboard Framework** | Streamlit |
| **Data Processing Tools** | Pandas, NumPy |